Course Code	**GEOTECHNICAL ENGINEERING**	L	T	P	C
23A01503T		3	0	0	3

Course Objectives:

The objectives of this course are to make the student to:

1. **Understand** the classification and compaction characteristics of different soil types and their engineering significance.
2. **Analyze** the concepts of effective stress, permeability, and seepage in soils and their impact on soil behavior.
3. **Apply** stress distribution theories and settlement computations to evaluate soil response under loads.
4. **Evaluate** shear strength properties of soil using various testing methods and their applications in geotechnical engineering.
5. **Assess** the stability of slopes using different analytical methods and suggest suitable slope protection measures.

Course Outcomes:

Upon successful completion of this course, students will be able to:

1. **Classify** soils based on their physical and index properties as per BIS and Unified classification systems.
2. **Analyze** soil permeability and seepage problems using Darcy's law and flow net concepts.
3. **Apply** stress distribution theories and settlement analysis to predict soil behavior under loading.
4. **Evaluate** shear strength of soils using experimental methods and interpret test results.
5. **Assess** slope stability and recommend suitable protection measures.

CO – PO Articulation Matrix:-

Course Outcomes	**PO 1**	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PSO1**	**PSO2**
CO -1	**3**	**2**	**2**	**1**	**-**	**-**	**-**	**-**	**-**	**2**	**-**	**2**	**3**	**2**
CO -2	**3**	**3**	**3**	**2**	**2**	**-**	**-**	**-**	**-**	**-**	**-**	**1**	**3**	**2**
CO -3	**3**	**2**	**2**	**2**	**2**	**-**	**-**	**-**	**-**	**-**	**-**	**1**	**2**	**2**
CO -4	**3**	**3**	**3**	**2**	**2**	**-**	**-**	**-**	**-**	**-**	**-**	**1**	**3**	**3**
CO -5	2	2	2	1	3	-	-	-	-	2	1	2	2	2

UNIT – I

INTRODUCTION: Soil Formation-Soil structure - Adsorbed water - Mass- Volume relationship — Relative density. Index Properties of Soils: Moisture Content, Specific Gravity, In-situ density, Grain size analysis — Sieve and Hydrometer methods — Consistency limits and indices -I.S. Classification of soils.

UNIT – II

PERMEABILITY: Soil Water — Capillary Rise — Flow of Water Through Soils - Darcy's Law-Permeability Factors Affecting — Laboratory Determination of Coefficient of Permeability-Permeability of Layered Systems.

SEEPAGE THROUGH SOILS: Total, Neutral and Effective Stresses - Quick Sand Condition — Seepage Through Soils — Flow Nets : Characteristics and Uses.

UNIT – III

STRESS DISTRIBUTION IN SOILS: Boussinesq's and Wester Guard's Theories For Point Loads and Areas Of Different Shapes — Newmark's Influence Chart Compaction: Mechanism of Compaction — Factors Affecting — Effects Of Compaction on Soil Properties. — Field Compaction Equipment — Compaction Control.

UNIT – IV

CONSOLIDATION: Types of Compressibility -Primary Consolidation and Secondary Consolidation - Stress History of Clay; E-P And E-Log P Curves — Normally Consolidated Soil, Over Consolidated Soil and Under Consolidated Soil — Pre-Consolidation Pressure and Its Determination — Terzaghi's I-D Consolidation Theory — Coefficient of Consolidation: Square Root Time and Logarithm of Time Fitting Methods.

UNIT – V

SHEAR STRENGTH OF SOILS: Importance of Shear Strength- Mohr's- Coulomb Failure Theories — Types of Laboratory Tests for Strength Parameters
Strength Tests Based on Drainage Conditions — Critical Void Ratio —Liquefaction.

TEXT BOOKS:

1. Soil Mechanics and Foundation Engg by K.R.Arora, Standard Publishers and DistributorsDelhi7thedition2009
2. Geotechnical Engineering by C.Venkataramiah, New Age International Pvt. Ltd,(2002).

REFRENCE BOOKS:

1. Soil Mechanics and Foundation by B.C.Punmia, Ashok Kumar Jain and Arun Kumar Jain, Laxmi, publications Pvt.Ltd., New Delhi17thedition 2017
2. Geotechnical Engineering by Iqbal H.Khan, PHI Pubilishers, 4th edition.
3. Basic and Applied Soil Mechanics by Gopal Ranjan & ASR Rao, New age International Pvt.Ltd, New Delhi 3rdedition 2016

Online Learning Resources:

https://nptel.ac.in/courses/105101201
https://nptel.ac.in/courses/105105185